

Part01

1. Why can't you override a sealed property or method?
Because sealed keyword stop inheritance on specific class and other Childs can't perform override on it.
2. What is the key difference between static and object members?
You call static members without instance, but object you need instance to call it.
3. Can you overload all operators in C#? Explain why or why not?
No not all operators can be overloaded like = cause it connected with the memory reference and conditional operators too cannot be overloaded cause they have short circuiting behavior with the compiler.
4. When should you consider changing the underlying type of an enum?
 1. Memory efficiency: the default take 4 bytes in memory but if you want to save memory you can use byte that take 1 byte.
 2. Need larger numbers: when you need the enum to hold larger number than the int
5. Why can't a static class have instance constructors?
Because it's not allowed to make objects from static classes.
6. Why is overriding ToString beneficial when working with custom classes?
You make the output of it meaningful and don't have to implement new method to print.
7. Can generics be constrained to specific types in C#? Provide an example.
Yes like class Helper<T> where T : IComparable.
8. What are the key differences between generic methods and generic classes?
In method only it being affected by generic type, but in the class the whole class can use the generic type.

9. Why might using a generic swap method be preferable to implementing custom methods for each type?

For code readability and reuse and reduce number of methods to make it easier to understand

10. Why is == not implemented by default for structs?

Because == is used for reference equality and struct is value type.